

EXPLORING THE MOVIE UNIVERSE WITH PYTHON: AN EDA JOURNEY



“Data Analyst”

Data Cleaning, Visualization & Insights from a Movie Dataset

Using Python - Pandas, Matplotlib & Seaborn





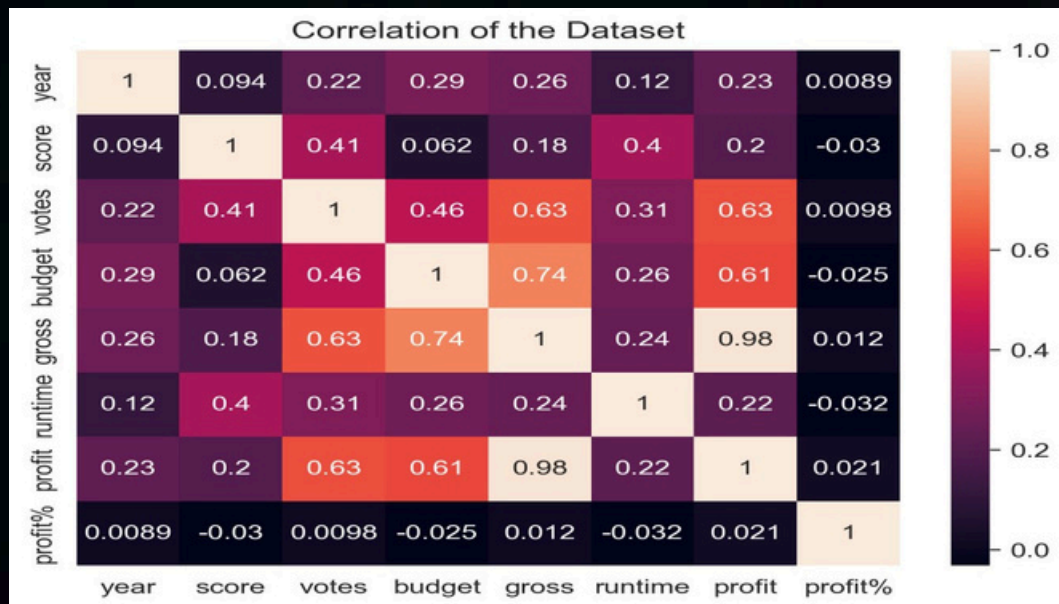
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...	name	rating	genre	year	score	votes	director	writer	star	country	budget	gross	company	runtime	profit	profit%
0	Avatar	PG-13	Action	2009	7.8	1100000	James Cameron	James Cameron	Sam Worthington	United States	237000000	2847246203	Twentieth Century Fox	162	2610246203	1101.369706
1	Avengers: Endgame	PG-13	Action	2019	8.4	903000	Anthony Russo	Christopher Markus	Robert Downey Jr.	United States	356000000	2797501328	Marvel Studios	181	2441501328	685.814980
2	Titanic	PG-13	Drama	1997	7.8	1100000	James Cameron	James Cameron	Leonardo DiCaprio	United States	200000000	2201647264	Twentieth Century Fox	194	2001647264	1000.823632
3	Star Wars: Episode VII - The Force Awakens	PG-13	Action	2015	7.8	876000	J.J. Abrams	Lawrence Kasdan	Daisy Ridley	United States	245000000	2069521700	Lucasfilm	138	1824521700	744.702735
4	Avengers: Infinity War	PG-13	Action	2018	8.4	897000	Anthony Russo	Christopher Markus	Robert Downey Jr.	United States	321000000	2048359754	Marvel Studios	149	1727359754	538.118303
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
7658	The Robinsons	N/A	Action	2020	0.0	0	Directors	Aleks Alifirenko Jr.	Billy Hartmann	United States	10000	20184964	Universal Pictures	90	20174964	201749.640000
7659	More to Life	N/A	Drama	2020	3.1	18	Joseph Ebanks	Joseph Ebanks	Shannon Bond	United States	7000	20184964	Universal Pictures	90	20177964	288256.628600
7660	Dream Round	N/A	Comedy	2020	4.7	36	Dusty Dukatz	Lisa Huston	Michael Saquella	United States	17000000	20184964	Cactus Blue Entertainment	90	3184964	18.735082
7661	It's Just Us	N/A	Drama	2020	0.0	0	James Randall	James Randall	Christina Roz	United States	15000	20184964	Universal Pictures	120	20169964	134466.426700
7662	Tee em el	N/A	Horror	2020	5.7	7	Pereko Mosia	Pereko Mosia	Siyabonga Mabaso	South Africa	8750000	20184964	PK 65 Films	102	11434964	130.685303
7663 rows × 16 columns																

This is the dataset which I used to do the EDA(Exploratory Data Analysis) and Data Cleaning work



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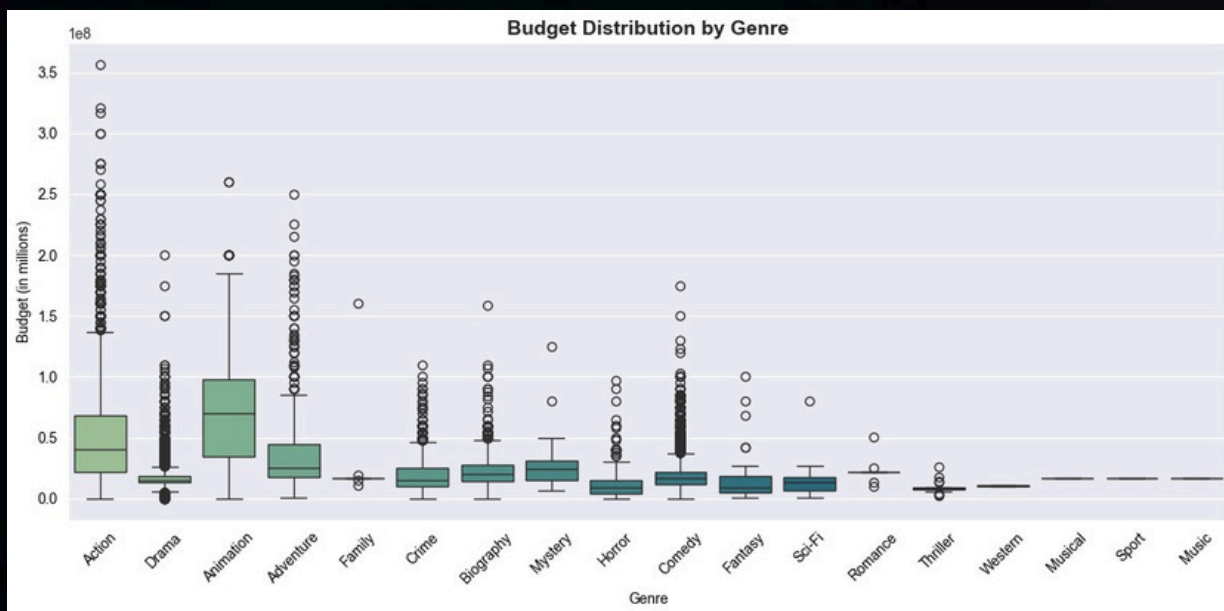


```
co_relation= df.corr(method='pearson',numeric_only=True)
sns.heatmap(co_relation,annot=True)
plt.title("Correlation of the Dataset")
plt.show()
plt.savefig("correlation.png", dpi=300, bbox_inches='tight')
```

Budget and profit show strong correlation with gross earnings. This suggests that higher investments often lead to higher returns—but not always. While Votes and runtime show weaker correlations, shows that audience engagement isn't solely driven by length or popularity metrics.



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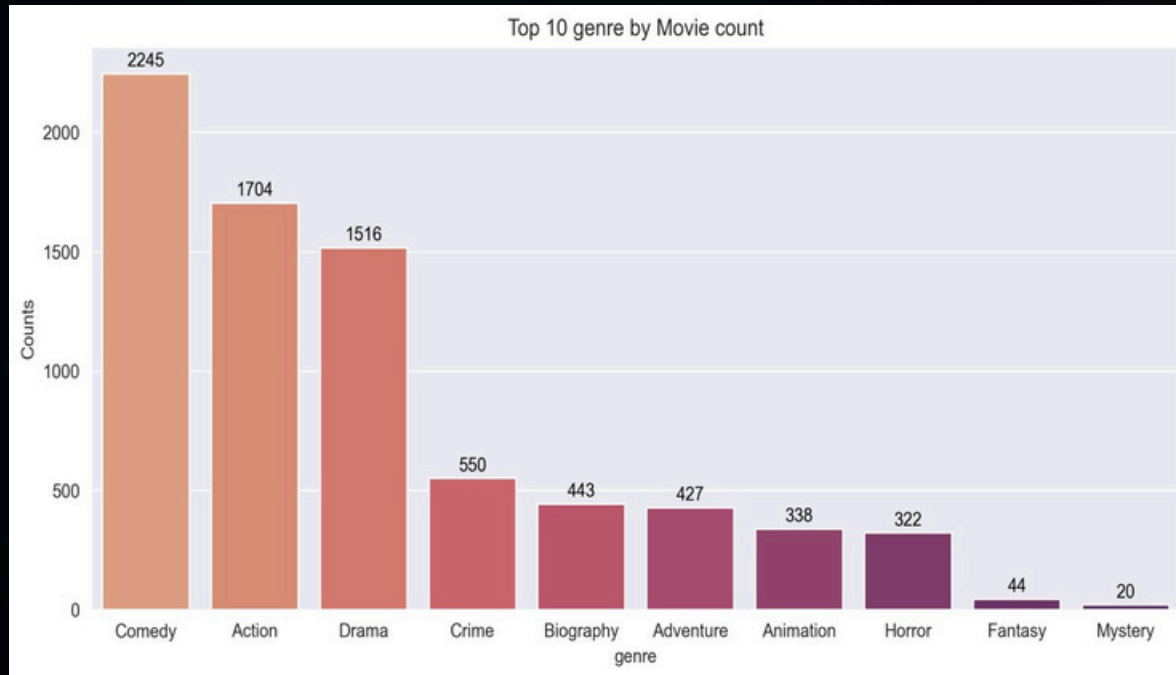


```
sns.set_style("darkgrid")
plt.figure(figsize=(12, 6))
sns.boxplot(x='genre', y='budget', data=df, palette='crest', hue='genre')
plt.xticks(rotation=45, color='black')
plt.yticks(color='black')
plt.title('Budget Distribution by Genre', fontsize=14, color='crimson')
plt.xlabel('Genre', color='black')
plt.ylabel('Budget (in millions)', color='black')
plt.tight_layout()
plt.show()
plt.savefig("Boxplot.png", dpi=300, bbox_inches='tight')
```

- Action and Adventure movies show the widest budget range from low-cost independent films to massive blockbusters highlighting their flexible production scale.
- Meanwhile, genres like Sports, Romance, and Musical tend to stay within smaller, more consistent budgets, reflecting predictable spending patterns



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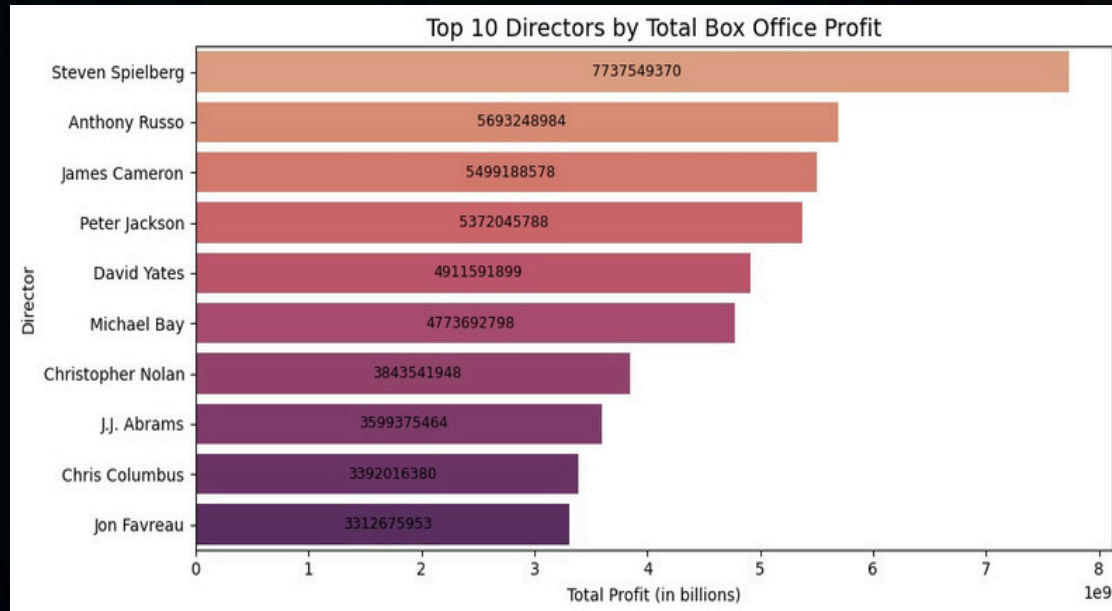


```
plt.figure(figsize=(10,5))  
ax = sns.barplot(x='genre', y='Counts', data=x,palette='flare', hue='genre')  
plt.title("Top 10 genre by Movie count")  
plt.tight_layout()  
for container in ax.containers:  
    ax.bar_label(container, fmt='%d', label_type='edge', padding=2, fontsize=10, color='black')  
plt.savefig("top_genres.png", dpi=300, bbox_inches='tight')
```

- Comedy leads the pack with the highest number of movies, showing its broad appeal across audiences.
- Fantasy and Mystery appear less frequently, suggesting they're more niche or selectively produced genres.



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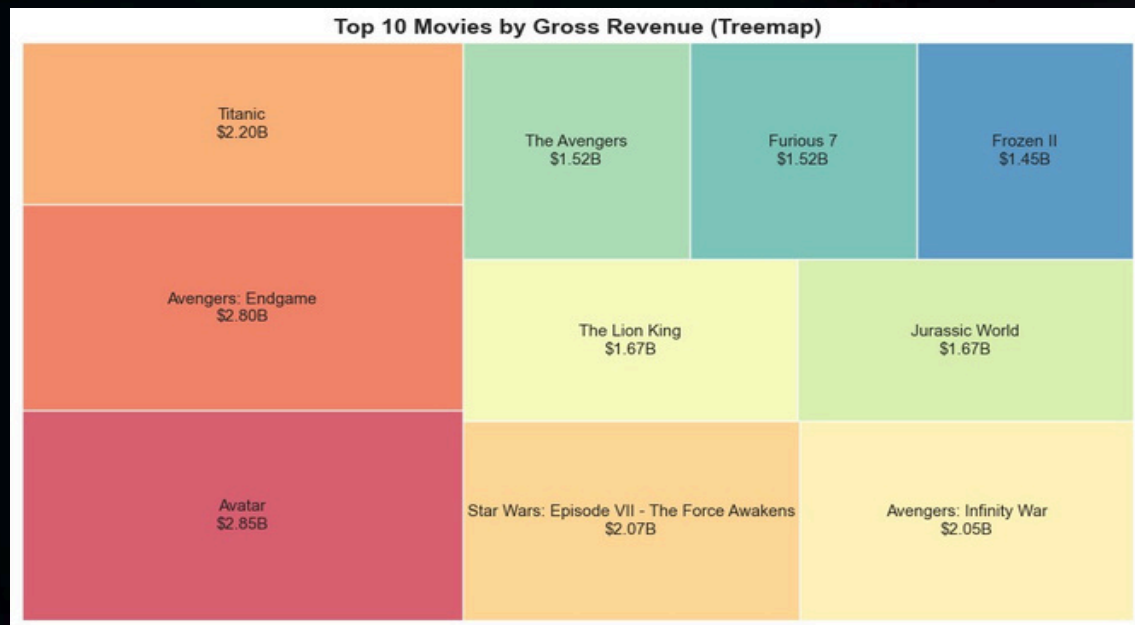
```
top_directors = df.groupby('director')['profit'].sum().nlargest(10).reset_index()

plt.figure(figsize=(10,5))
ax = sns.barplot(x='profit', y='director', data=top_directors, palette='flare')
plt.title("Top 10 Directors by Total Box Office Profit", fontsize=14)
plt.xlabel("Total Profit (in billions)")
plt.ylabel("Director")
for container in ax.containers:
    ax.bar_label(container, fmt='%.0f', label_type='center', padding=3, fontsize=9, color='black')
plt.tight_layout()
plt.show()
plt.savefig("top10_directors.png", dpi=300, bbox_inches='tight')
```

- Steven Spielberg leads the highest total box office profit, showcasing his consistent success across blockbuster hits.
 - Christopher Nolan one of my favorite director is also in the list
- The chart highlights how a director's vision and reputation can significantly influence a film's financial performance



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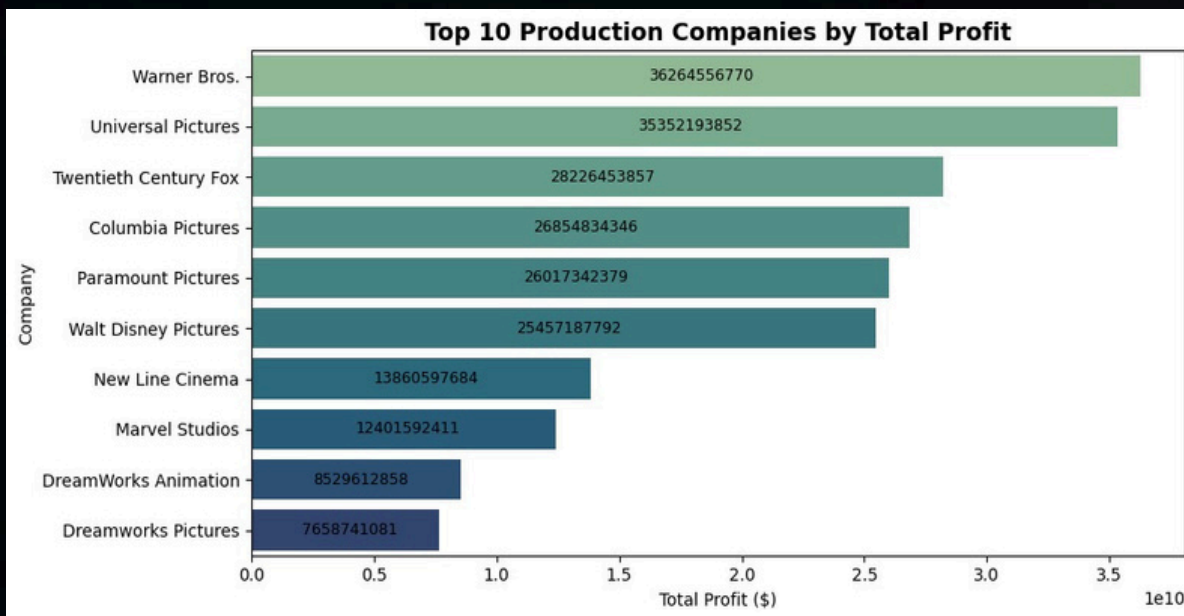


```
import squarify
top10 = df.sort_values(by='gross', ascending=False).head(10)
labels = [f"{name}\n${gross/1e9:.2f}B" for name,
          gross in zip(top10['name'],
                       top10['gross'])]
plt.figure(figsize=(12,7))
squarify.plot(
    sizes=top10['gross'],
    label=labels,
    alpha=0.8,
    color=sns.color_palette('Spectral', 10)
)
plt.title('Top 10 Movies by Gross Revenue (Treemap)',
          fontsize=14, weight='bold')
plt.axis('off')
plt.show()
plt.savefig("top10movies.png", dpi=300, bbox_inches='tight')
```

- Avatar remains the highest-grossing movie of all time, even after 16 years—an unmatched milestone in cinematic history.
- The tree map shows how a few standout titles dominate global box office revenue



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```
company_profit = df.groupby('company')['profit'].sum().sort_values(ascending=False).head(10)
plt.figure(figsize=(10,5))
ax = sns.barplot(
    y='company',
    x='Total Profit',
    data=company_profit,
    palette='crest'
)
plt.title('Top 10 Production Companies by Total Profit',
         fontsize=14, weight='bold')
plt.xlabel('Total Profit ($)')
plt.ylabel('Company')

for container in ax.containers:
    ax.bar_label(container, fmt='%.0f', label_type='center',
                 padding=3, fontsize=9, color='black')

plt.tight_layout()
plt.show()
```

- Warner Bros tops the profit chart, followed closely by Universal Pictures highlighting the dominance of major studios in global box office success.
- The chart reflects how production scale and franchise power drive profitability in the film industry



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```
avg_profit_by_year = df.groupby('year')['profit'].mean().reset_index()
plt.figure(figsize=(10,5))
sns.lineplot(data=avg_profit_by_year, x='year', y='profit', marker='o', color='gold')
plt.title('Average Profit Over Years', fontsize=14, weight='bold')
plt.xlabel('Year')
plt.ylabel('Average Profit')
plt.tight_layout()
plt.show()
plt.savefig("Averageprofit.png", dpi=300, bbox_inches='tight')
```

- The chart reveals a steady rise in average movie profits from the 1980s to the mid-2020s, reflecting the growing scale of film production and global box office reach.
- This upward trend highlights how the industry has evolved driven by franchise power, international markets, and advancements in cinematic technology.



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What I have done as a Data Analyst from this dataset is

- Identified data quality issues and “**Inconsistencies**”
- Optimized “**Data Types**” for efficient analysis
- Engineered “**New Features**” to enrich insights
- Conducted “Exploratory Data Analysis”(EDA) to uncover trends

This journey shows how data can uncover hidden patterns in “storytelling,” “production”, and” audience impact” transforming raw numbers into powerful narratives

This project reflects my passion for turning data into stories and my journey as an aspiring “**Data Analyst**”.

Explore the full project: https://github.com/Akshay7368/Datacleaning_work